

Point-by-Point Response to Reviewers

Scientific Reports Manuscript Revision: *Ontology-constrained multi-LLM scoring of hypothesis support in the predictive processing literature*

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We express our sincere gratitude to the Editor and Reviewer #1 for their constructive, rigorous, and insightful evaluation of our manuscript. In response, we have conducted targeted sensitivity analyses, empirical ablations, and statistical clarifications. In accordance with the Editor’s requirement for computational workflows, all underlying pipeline and analysis code, ontological specifications, and reproduction scripts have been permanently deposited in a DOI-assigning repository on Zenodo (<https://doi.org/10.5281/zenodo.22182632>; GitHub: <https://github.com/HNXJ/mllm-public>), and an explicit Code Availability section is provided in the manuscript. All reported values correspond strictly to immutable, audited experimental tables.

Below, we address the reviewer’s concerns point by point.

Response to Reviewer #1

Major Concern 1: Verification of model reasoning and human expert concordance.

Author Response: We thank the Reviewer for raising this critical question regarding the validity of autonomous LLM evaluations compared to expert human judgment. In response, we expanded our domain validation framework, established the independent scientific paper as the primary inferential unit, and evaluated both systematic magnitude calibration and rank correspondence across the three core predictive coding hypotheses (Supplementary Table S14 and Supplementary Fig. S2):

1. **Paper-Level Hypothesis Validation ($K = 2$ Domain Consensus):** To evaluate autonomous Council scoring against independent domain-expert judgment without confounding multi-rater consensus with single-expert evaluations, we conducted paired paper-level analyses using the subset of independent papers where two computational neuroscientists ($K = 2$) independently evaluated mechanisms within each specific hypothesis. Across all six evaluated conditions, human consensus means consistently exceeded Council means:

- *Local Oddball:* H_1 (Predictive Suppression): Human $+0.537 \pm 0.041$ vs. Council $+0.337 \pm 0.027$ ($N = 21, t(20) = 4.69, p = 0.0001$); H_2 (Prediction Errors): Human $+0.552 \pm 0.045$ vs. Council $+0.363 \pm 0.026$ ($N = 23, t(22) = 5.42, p < 0.0001$); H_3 (Ubiquity): Human $+0.520 \pm 0.034$ vs. Council $+0.268 \pm 0.045$ ($N = 14, t(13) = 5.95, p < 0.0001$).

- *Global Oddball*: H_1 : Human $+0.600 \pm 0.048$ vs. Council $+0.299 \pm 0.031$ ($N = 18, t(17) = 5.32, p < 0.0001$); H_2 : Human $+0.560 \pm 0.043$ vs. Council $+0.259 \pm 0.049$ ($N = 22, t(21) = 6.54, p < 0.0001$); H_3 : Human $+0.557 \pm 0.065$ vs. Council $+0.312 \pm 0.057$ ($N = 11, t(10) = 3.32, p = 0.0077$).

These statistically detectable paired mean differences reflect systematic score attenuation and conservative evidence weighting by autonomous models relative to human domain experts (individual paired effect estimates and 95% CIs are reported in Supplementary Table S14). In a sensitivity analysis incorporating all available human ratings ($N \leq 31$), all six paired differences remained in the same direction with $p < 0.05$.

Across-paper rank correspondence was heterogeneous across hypotheses (Supplementary Table S14): moderate rank correspondence was observed for Feedforward Prediction Errors in both contexts (LO- H_2 : Spearman $\rho = 0.612, p = 0.0019$; GO- H_2 : $\rho = 0.492, p = 0.020$), whereas rank correspondence was lower or unaligned for Predictive Suppression (LO- H_1 : $\rho = 0.196, p = 0.396$; GO- H_1 : $\rho = -0.136, p = 0.590$) and Ubiquity (LO- H_3 : $\rho = 0.467, p = 0.092$; GO- H_3 : $\rho = 0.218, p = 0.519$).

2. **Manual Source-Evidence Check**: To test whether autonomous model scores were grounded in primary text rather than confabulation, we performed a manual source-evidence check across 27 prespecified Human–Council disagreement cases (defined as one ontology-factor score for one paper and context; Supplementary Table S8). A human reviewer returned to the original paper and judged whether the cited source passage supported the Council-assigned score. This check verified that 77.8% (21/27) of Council assignments were directly supported by primary-source text.

Major Concern 2: Sensitivity to stochastic decoding temperature, repeated sampling, and model architecture variability.

Author Response: To directly separate stochastic sampling noise (temperature and repeated generations) from genuine model architecture differences, we conducted an 837-evaluation factorial design ($31 \text{ papers} \times 3 \text{ models} \times 3 \text{ temperatures } [T \in \{0.00, 0.35, 0.70\}] \times 3 \text{ repeats}$; Supplementary Table S9 and Fig. S3):

1. **Operational Completion by Architecture**: The tested models differed substantially in structured generation stability. In our primary evaluation model (**gemma-4-31b-it**), all 279/279 (100.0%) inference attempts completed all 72 canonical factor slots (yielding 6,054 evaluated factor cells). By contrast, in reasoning models with extensive chains-of-thought, long intermediate traces frequently reached fixed token generation limits before serializing final outputs: **phi-4-reasoning-plus** achieved complete evaluations in 130/279 (46.6%) attempts (1,449 factor cells), and **olmo-3-32b-think** completed 30/279 (10.8%) attempts under local context constraints. Consequently, we base primary stochastic stability conclusions on **gemma-4-31b-it** and report reasoning models as sensitivity explorations.
2. **Cross-Temperature Score Stability**: Within the primary model (**gemma-4-31b-it**, $N = 679$ overlapping factor cells), repeat-averaged scores were highly stable across decoding temperatures: comparing $T = 0.00$ to $T = 0.35$ yielded Pearson $r = 0.9900$ (MAE = 0.0266), and comparing $T = 0.00$ to $T = 0.70$ yielded $r = 0.9760$ (MAE = 0.0421) across the $[-1.0, +1.0]$ ontology space. Among completed evaluations in **phi-4-reasoning-plus** ($N = 104$ cells), temperature agreement was moderate ($r = 0.6505$, MAE = 0.1421 between $T = 0.00$ and $T = 0.70$).
3. **Within-Temperature Replicate Reliability**: Pairwise agreement across independent stochas-

tic generations held at fixed temperatures was similarly high in the primary model, though reliability declined slightly with increasing temperature: mean replicate correlation and error were $\bar{r}_{\text{rep}} = 0.9905$ (MAE = 0.0126) at $T = 0.00$ ($N = 639$ triples), $\bar{r}_{\text{rep}} = 0.9724$ (MAE = 0.0384) at $T = 0.35$ ($N = 619$ triples), and $\bar{r}_{\text{rep}} = 0.9444$ (MAE = 0.0585) at $T = 0.70$ ($N = 572$ triples).

4. **Architectural Nuance vs. Sampling Noise:** These findings demonstrate that stochastic sampling noise within a model architecture is small at low temperatures and increases moderately at $T = 0.70$. However, model-to-model differences in hypothesis weighting remain measurable, motivating multi-model ensemble aggregation to balance architecture-dependent differences across models.

Major Concern 3: Corpus composition, subgroup heterogeneity, and evidence grounding controls.

Author Response: We performed a comprehensive subgroup sensitivity analysis across Study Type, Modality, Era, and Publication Status (Supplementary Table S12), alongside controlled evidence ablations (Supplementary Table S10 and Fig. S4):

- **Subgroup Heterogeneity & Publication Status:** Rather than uniform invariance, the local-to-global displacement is heterogeneous across corpus strata: Human In Vivo studies ($N = 7$) showed a negative displacement ($\Delta_{\text{LO-GO}} = -0.059$), whereas Animal In Vivo electrophysiology ($N = 5$) exhibited a positive displacement ($\Delta_{\text{LO-GO}} = +0.087$). Studies from 1982–2014 ($N = 12$) and 2015–2020 ($N = 7$) exhibited negative displacements (-0.024 and -0.041), whereas recent studies (2021–2025, $N = 12$) exhibited positive displacement ($+0.038$). In the working corpus, 30 studies were peer-reviewed journal publications and 1 was a preprint (Westerberg & Xiong 2025, $\Delta_{\text{LO-GO}} = -0.400$).
- **Evidence Grounding Control:** Across 54 inference calls, when evidence text was shuffled (Negative Control H_0), context displacement collapsed to $\Delta_{\text{LO-GO}} = -0.004$ (compared to $+0.038$ for intact text), and under prior-only prompts without text, displacement collapsed to $\Delta_{\text{LO-GO}} = +0.000$ (Supplementary Table S10 and Fig. S4), demonstrating that scoring is sensitive to the supplied empirical text.

Major Concern 4: Statistical reporting, clustering, and outlier sensitivity.

Author Response: We audited all manuscript statistical assertions against paper-clustered inferential standards:

- We explicitly state in Methods 4.11 that main-text comparisons are framed descriptively and that non-parametric permutation reference distributions use $B = 10,000$ iterations.
- Across $N = 28$ paired empirical papers, the mean paired LO–GO difference was -0.0177 ($t(27) = -0.361, p = 0.7208, 95\% \text{ CI} = [-0.1183, 0.0829]$; Wilcoxon $W = 24.0, p = 0.0736$). After excluding Westerberg & Xiong (2025) in the prespecified single-study sensitivity analysis, the mean paired LO–GO difference was $+0.0297$ ($t(26) = 2.302, p = 0.0296, 95\% \text{ CI} = [0.0032, 0.0563]$; Wilcoxon $W = 10.0, p = 0.0131$).
- We distinguish descriptive factor-cell reliabilities from independent paper-level inferences to prevent pseudo-replication.

Minor Concern 1: Westerberg citation and year normalization.

Author Response: We normalized all bibliographic citations and in-text references to Westerberg & Xiong (2025) consistently throughout the manuscript and Supplementary Information, accurately classifying its publication status as a preprint.

Minor Concern 2: Figure description accuracy and figure-text-removal sensitivity.

Author Response: To evaluate whether findings depend on multimodal figure captions extracted via DeepRead:

- We performed a 10-figure manual expert validation against primary source PDFs, achieving a 10/10 overall PASS (directional concordance: 10/10; ontology mapping: 9 correct, 1 partial; numerical extraction: 9 correct, 1 minor error).
- In a full 31-paper figure-text ablation sweep (533 paired factor cells), removing figure-derived text blocks preserved overall factor directionality (directional concordance = 99.42% on non-zero cells [518/521], all-cell sign agreement = 97.19% [518/533], Pearson $r = 0.8539$) while producing measurable paper-level score adjustments (MAE = 0.2253; Supplementary Table S11 and Fig. S5), with Westerberg & Xiong (2025) remaining a focal divergent outlier in both conditions (Rank 28/31 Full vs Rank 29/31 NoFig).

Minor Concern 3: Specification of decoding parameters and sampling reproducibility.

Author Response: We explicitly documented all decoding parameters across inference conditions (Supplementary Table S6 and Methods Section 4.3). Specifically, we verified that varying sampling temperatures across $T \in \{0.00, 0.35, 0.70\}$ and running independent stochastic repeats yields stable factor scores in the primary evaluation model ($r \geq 0.9760$, MAE ≤ 0.0421 ; Supplementary Table S9 and Fig. S3).